

Joint Training with Machine-Generated Text Detection Can Distort Sarcasm Detection: A Bilingual Diagnosis of Class-Conditional Negative Transfer

Anonymous ACL submission

Abstract

A sarcastic utterance can communicate an attitude that differs from what its literal wording suggests. Because recognizing this contrast depends on implicit intent and context, sarcasm is difficult to detect automatically. LLM-generated examples could improve detection by augmenting training data. However, this strategy requires LLMs to produce text that both conveys sarcasm and resembles human writing, so sarcasm and authorship must be evaluated together. Although sarcasm detection and machine-generated text detection (MGTD) are usually studied separately, their interaction in shared representations remains insufficiently understood. MGTD must encode source cues, whereas sarcasm detection should reflect intent rather than source. To test whether any resulting interference is confined to one language, we provide a bilingual diagnosis of joint MGTD–sarcasm training in Chinese and English. We harmonize bilingual resources, generate topic-matched AI text, and build DualTest with source and sarcasm labels, defining AI labels by requested style. Compared with their single-task counterparts, jointly trained models are markedly more likely to misclassify AI-generated texts prompted to be non-sarcastic as sarcastic. Detector-based human-authorship scores and LLM ratings of human-likeness also produce different rankings of the outputs. These findings motivate subgroup evaluation and caution against detector-only assessment.

1 Introduction

In a sarcastic sentence, the speaker’s attitude or intent may differ from what the literal wording suggests. As a characteristically human form of expression, sarcasm relies on implicit contrast, intent, and context, making it difficult to detect automatically. Detection errors can reverse sentiment predictions and disrupt opinion-mining or content-moderation systems (Tay et al., 2018; Vidgen et al.,

2019). Annotation outcomes also vary with elicitation method and annotator perspective (Oprea and Magdy, 2020; Casola et al., 2024), while model behavior is sensitive to domain, style, and language (Jang and Frassinelli, 2024; Casola et al., 2024) (§2.1).

One potential response is to augment training data with LLM-generated examples. This strategy is useful only if generated text both conveys the intended sarcasm and resembles human writing; otherwise, source-specific artifacts may become shortcuts. Sarcasm and authorship must therefore be considered together. Human-authored and AI-generated texts can each be sarcastic or non-sarcastic, yielding all four combinations.

Machine-generated text detection (MGTD) imposes a contrasting representational demand: it must distinguish human-written from AI-generated text and thus favors source-predictive representations (Guo et al., 2023; Wang et al., 2024) (§2.2). These stylistic or distributional cues may overlap with features used for sarcasm detection. The labels do not conflict; the potential conflict lies in how the tasks organize a shared representation. MGTD must encode source, whereas sarcasm detection should reflect intent rather than source. Shared-encoder multi-task learning can suffer task conflict and negative transfer by exploiting information useful for one task but spurious for another (Mueller et al., 2022; Hu et al., 2022b). Separate models avoid this coupling but forgo efficient encoder sharing, motivating a direct test of whether joint training contaminates sarcasm decisions.

Detector scores also matter downstream because they can become generation rewards. ViSP derives PPO rewards from a sarcasm detector (Wang et al., 2026). Yet useful generated sarcasm must satisfy two distinct objectives: conveying sarcastic intent and being perceived as human-like. A dual-head detector may appear to provide corresponding sig-

nals through sarcasm and authorship probabilities, but authorship probability is not itself a measure of human-likeness. We therefore test whether these detector scores track source-blinded LLM ratings of sarcasm and perceived human-likeness (§6).

To test these risks rather than assume MGTD is a beneficial auxiliary task, we compare same-architecture dual- and single-head models on bilingual Chinese–English data and build DualTest for source $\times$ sarcasm subgroup analysis. The evaluated joint-training procedure sharply increases the sarcasm false-positive rate on AI text prompted to be non-sarcastic; we call this class-conditional negative transfer. The same pattern persists when the AI test texts are regenerated with Qwen3.5-9B, while the RCNN and interaction-projection configurations show lower rates than the plain configuration.

We then examine the shared representation. Source information is highly decodable and correlates with sarcasm decisions. To connect encoded information to actual decisions beyond probe accuracy (Hewitt and Liang, 2019; Elazar et al., 2021), we combine probes, task-head correlations, and representation interventions. These analyses identify a non-random high-variance direction to which the subgroup gap is sensitive, supporting a source-correlated shortcut hypothesis and providing a concrete target for causal analysis (§5.3.2, Appendix E).

Contributions. To our knowledge, we provide the first bilingual diagnostic study of how joint training with MGTD changes sarcasm detection across Chinese and English, together with DualTest for source $\times$ sarcasm analysis. Same-architecture comparisons across models and generators identify class-conditional negative transfer concentrated on AI-generated text prompted to be non-sarcastic; cross-architecture comparisons characterize how its severity varies across the configurations. Representation analyses support the source-correlated shortcut hypothesis and define concrete targets for causal analysis. Finally, an exploratory detector–judge comparison illustrates a downstream risk of using detector scores to evaluate generated sarcasm and perceived human-likeness.

2 Related Work

2.1 Sarcasm Detection and Multilingual Resources

Feature-engineered methods (Riloff et al., 2013), neural sequence models (Ghosh and Veale, 2016; Tay et al., 2018), and pretrained language models (Khatri and P, 2020; Du et al., 2022) have each advanced sarcasm detection. The task remains difficult in multilingual and class-imbalanced settings, especially because datasets differ in whether labels reflect intended sarcasm or annotator-perceived sarcasm (Oprea and Magdy, 2020; Abu Farha et al., 2022; Casola et al., 2024). Although scores are not directly comparable across datasets, languages, modalities, or evaluation setups, they illustrate substantial benchmark variability. On the English iSarcasmEval benchmark (Abu Farha et al., 2022), the top two systems report positive-class F1 of 0.605 and 0.569; across MultiPiCo languages (Casola et al., 2024), zero-shot ChatGPT positive-class F1 ranges from 0.152 to 0.567. On the bilingual multimodal SarcNet benchmark (Yue et al., 2024), a text-only XLM-R baseline reaches 0.734 F1.

RCNNs combine recurrent context with convolutional pooling for text classification (Lai et al., 2015) and have also been evaluated for multilingual sarcasm detection (Yang and Ikeda, 2024); we therefore include an RCNN intermediate configuration in §4.1.

The benchmark results summarized above come from different datasets, so §5.1 provides directly comparable mBERT dual-head baselines on DualTest. None of the sarcasm resources reviewed above provides both sarcasm and human-versus-AI source labels. DualTest supplies both labels in a bilingual Chinese–English setting.

The Chinese and English source datasets differ in domain, annotation procedure, and class balance, so bilinguality introduces additional distributional variation. We treat language as a robustness and stratification dimension rather than as the primary transfer objective: the study asks whether the subgroup failure is confined to one language, not whether knowledge transfers across languages. We therefore report language-stratified results in addition to pooled metrics.

2.2 Machine-Generated Text Detection and Authorship Verification

MGTD comprises zero-shot model-likelihood methods, such as probability-curvature detectors (Mitchell et al., 2023), and discriminative models trained on paired human–AI corpora (HC3; Guo et al., 2023). Generalization is a central difficulty: M4 (Wang et al., 2024) provides a benchmark across generators, domains, and languages and shows that detectors struggle on unseen domains or LLMs, while RAID (Dugan et al., 2024) stress-tests detectors under unseen generators, decoding variations, and adversarial attacks. This generalization gap is consistent with reliance on source-predictive regularities that are partly generator- or domain-specific; such cues are relevant here because they may enter a representation shared with sarcasm detection. Bevendorff et al. (2025) distinguish closed-set authorship attribution from open-set authorship verification and argue that realistic LLM detection is better cast as the latter. Our binary MGTD setup is narrower: we use it as a controlled source-classification auxiliary task, not as a measure of text quality.

In our literature search, we found no work that jointly trains sarcasm detection and MGTD with a shared encoder.¹ This gap motivates our study because source information has different roles in the two tasks: it is the target of MGTD but should not by itself determine a sarcasm detector’s prediction. Our study tests whether a shared representation preserves that distinction.

2.3 Multi-Task Negative Transfer, Shortcut Learning, and Shared-Encoder Mitigation

In multi-task learning, auxiliary tasks are not always beneficial: transfer depends on task relatedness and alignment between their data distributions (Zhang et al., 2023; Wu et al., 2020), while gradient conflict is a central optimization challenge that can contribute to negative transfer (Yu et al., 2020; Liu et al., 2021). Mueller et al. (2022) provide a relevant NLP precedent through matched single- and multi-task comparisons. They find similar negative-transfer patterns in canonical and text-

to-text systems and note that models may infer task identity from input distributions, a strategy that may not transfer reliably out of distribution. He and Choi (2021) and Li et al. (2023) further analyze functional specialization in multi-task Transformer encoders. Shortcut learning offers a complementary reading: models may exploit features that are predictive in the training distribution yet fail under distribution shift (Geirhos et al., 2020; Ye et al., 2026). In MTL, shared representations can increase reliance on spurious correlations (Hu et al., 2022b; Chai et al., 2025). Related multi-dataset work shows that adding data can introduce spurious correlations and reduce worst-group performance (Compton et al., 2023).

Shared-encoder mitigation either isolates task-specific parameters (Pfeiffer et al., 2021, 2020; Karimi Mahabadi et al., 2021; Stickland and Murray, 2019) or modifies information flow within a shared computation graph (Wu et al., 2019; Pilault et al., 2021). Our analysis is closer to the second design direction: we compare a no-module baseline with two intermediate-representation variants and test whether degradation changes across source $\times$ sarcasm subgroups (§5.3).

2.4 Misalignment in Detector-Based Evaluation and Rewards

The representational conflict studied above may also matter when detector outputs are reused beyond classification, particularly as evaluation or reward signals for generated text. Automated scorers, including LLM judges, are increasingly used to evaluate generated text (Zheng et al., 2023). Model-based scores are also optimized as rewards, but overoptimization can increase a proxy score while degrading a gold-standard measure (Gao et al., 2023). A detector can achieve useful classification accuracy without its continuous output being calibrated as a measure of style strength or generation quality. Multimodal sarcasm generation may face this risk: ViSP (Wang et al., 2026) uses the DIP sarcasm detector (Wen et al., 2023) as a PPO reward (Schulman et al., 2017). As an exploratory downstream analysis, Section 6 compares detector scores for human authorship with separately obtained, source-blinded LLM ratings of perceived human-likeness.

¹Searches conducted in May 2026 and updated in July 2026 across Semantic Scholar, arXiv, ACL Anthology, and Google Scholar, using keyword combinations including “sarcasm detection,” “machine-generated text detection,” “AI-generated text detection,” “multi-task learning,” and “joint detection.”

3 Dataset Construction

Testing for interference requires separate task supervision and evaluation labels for authorship and sarcasm. Here “source” denotes human-versus-AI authorship; “corpus” denotes an input dataset. We combine iSarcasm (Oprea and Magdy, 2020; Abu Farha et al., 2022), CSC (Jang and Frassinelli, 2024; Jang et al., 2023), ToSarcasm (Liang et al., 2022), News Headlines v2 (Misra and Arora, 2023; Misra and Grover, 2021), and the Open Chinese Internet Sarcasm Corpus (Zhu, 2022). None of these datasets provides both labels. Mapping their annotations to a common binary sarcasm label yields 44,862 pooled instances (approximately 87% English), from which we construct three training sets and DualTest (Appendix A).

Sarcasm classification training set, SarcTrain (10,000, all human). Each language $\times$ label bucket contains 2,500 instances. Three buckets are downsampled; the Chinese sarcastic bucket starts from 2,396 records and adds 104 oversampled records. The resulting set is organized into five folds.

MGTD training set, AuthTrain (19,995). AuthTrain draws 9,999 approximately language-balanced human seeds from SarcTrain (5,000 English / 4,999 Chinese) after seed-level generation quality control excludes one Chinese seed. The Ollama model tagged deepseek-r1:8b (DeepSeek-AI, 2025; Ollama, 2025) produces 9,996 retained AI counterparts; three selected human seeds remain unmatched, giving 99.97% pairing coverage. Appendix A details the exclusions.

We use four language $\times$ style templates, with each seed’s label determining the requested style. Authorship and requested-style labels form two separately controlled, approximately balanced factors, enabling source $\times$ sarcasm subgroup analysis (Appendix A).

Generation training set, GenTrain (2,428; used in §6). From 5,257 eligible positive-sarcasm records in iSarcasm and ToSarcasm, we sample 2,103 records (40%; 1,214 English and 889 Chinese). Chinese oversampling adds 325 records, yielding 1,214 per language. The resulting set contains 374 rewriting, 839 response, and 1,215 comment instances and serves as the instruction-tuning data for supervised fine-tuning (SFT; Appendix A).

Diagnostic test set, DualTest (4,664). The human subset combines official iSarcasm (1,400 English) and ToSarcasm (972 Chinese) test sets. After generation quality control, AI counterparts remain for 2,292 of 2,372 texts (96.6% coverage). Human texts retain their annotations; for AI texts, the sarcasm label records the prompted condition rather than an independently annotated judgment of perceived sarcasm.

The four source $\times$ sarcasm groups contain 823 human-sarcastic, 1,549 human-non-sarcastic, 779 AI-sarcastic, and 1,513 AI-non-sarcastic instances. Appendix A gives the language $\times$ source $\times$ sarcasm distribution.

4 Dual-Head Detection Model

4.1 Tasks and Architecture

Given a Chinese or English input text x , the model produces two separate, non-mutually-exclusive binary predictions: whether x is labeled sarcastic and whether its source label is human or AI. Any combination of the two labels is valid. The tasks share a single DeBERTa-v3-large encoder (He et al., 2023), allowing us to test how incorporating MGTD into joint training changes the representation read by the sarcasm head. Encoder outputs are aggregated by an intermediate module and then passed to two task-specific classification heads; Figure 1 shows the RCNN configuration.

The RCNN configuration uses BiLSTM + CNN as its intermediate module. A single-layer bidirectional LSTM with hidden size 256 per direction produces a 512-dimensional output at each token position. We concatenate this output with the 1,024-dimensional encoder output, then apply three groups of one-dimensional convolutions with kernel widths 3/4/5 and 128 channels each. Global max pooling produces a 384-dimensional vector z . After dropout, z is passed to the two heads, yielding a DeBERTa-based dual-head RCNN configuration.

To test the intermediate layer’s role, §5 compares two variants. The plain model sends [CLS] directly to both heads. The interaction-projection model forms two learned 256-dimensional projections, h_s and h_a , and updates them as $h'_s = h_s + P_{a \rightarrow s}(h_a)$ and $h'_a = h_a + P_{s \rightarrow a}(h_s)$. Because each branch supplies only one key-value vector, every attention head assigns a weight of one to that vector; the module therefore reduces to residual cross-branch affine projections, where $P_{a \rightarrow s}$

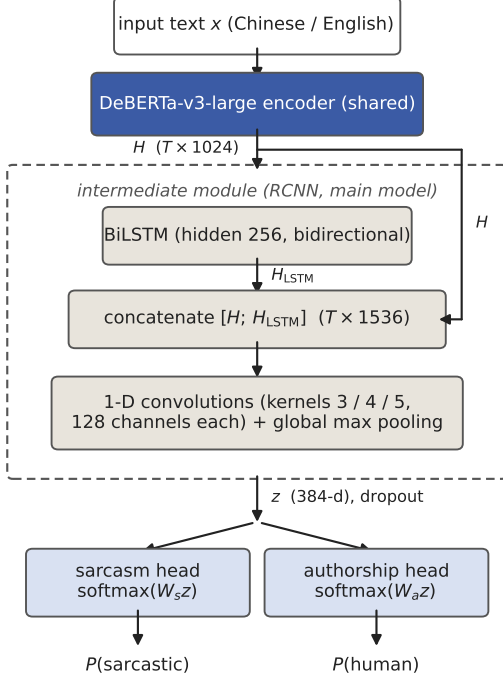


Figure 1: RCNN dual-head configuration: a shared DeBERTa-v3-large encoder, an RCNN intermediate module (dashed), and two task-specific heads. The comparisons in §5 remove this component in the plain configuration or replace it with a task-branch interaction-projection module, while retaining the same encoder backbone and task definitions (Appendix B).

and $P_{s \rightarrow a}$ each compose the attention value and output projections. The configurations retain the same encoder and task definitions while testing distinct intermediate-representation designs (Appendix B).

4.2 Training Protocol

The training protocol supports controlled dual-single comparisons in the plain and RCNN configurations. Within each comparison, the models use the same encoder architecture, intermediate configuration, sarcasm data and fold assignments, and sarcasm-specific hyperparameters. The dual model additionally receives AuthTrain batches, updates shared parameters from both tasks, follows the mixed-task schedule, and uses joint early stopping; the comparison therefore evaluates the complete joint-training procedure rather than any single component.

Training uses a mixed loader that supplies task-homogeneous batches according to a 2:1 MGTD:sarcasm schedule, matching the approxi-

mate data-size ratio between AuthTrain and SarcTrain. Let θ_{sh} denote parameters updated by both tasks, let $z_s(x)$ and $z_a(x)$ be the representations read by the sarcasm and MGTD heads, and let W_s and W_a denote the two classification heads. The two representations coincide in the plain and RCNN configurations; in the interaction-projection configuration, they are the task-specific h'_s and h'_a defined above. At step t , the loss is computed only for the task corresponding to the current batch B_t :

$$\mathcal{L}(B_t) = \begin{cases} \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}_w(W_s z_s(x), y), & B_t \subset \text{SarcTrain}, \\ \frac{1}{|B_t|} \sum_{(x,y) \in B_t} \text{CE}(W_a z_a(x), y), & B_t \subset \text{AuthTrain}, \end{cases} \quad (1)$$

where CE_w is a pre-specified cost-sensitive cross-entropy for sarcasm (weights are provided in Appendix B). For the class-balanced SarcTrain dataset, the weighted loss assigns a greater penalty to errors on sarcastic examples. Softmax is applied to logits only when probabilities are required. The gradient flow follows from the equation: on sarcasm batches, $\partial \mathcal{L} / \partial W_a = 0$ (symmetrically, $\partial \mathcal{L} / \partial W_s = 0$ on MGTD batches), whereas θ_{sh} receives gradients from both types of batches. Unless otherwise stated, binary predictions use a fixed probability threshold of 0.5 rather than a threshold tuned on DualTest.

The encoder uses a lower learning rate than newly added modules. The sarcasm-head rate is fixed across configurations; MGTD-head rates are reported in Appendix B.

Model selection accounts for both tasks. At epoch e , the early-stopping signal is the equally weighted sum of the two validation losses, each normalized by its first-epoch value:

$$\mathcal{L}_{\text{stop}}^{(e)} = \frac{1}{2} \frac{\mathcal{L}_s^{(e)}}{\mathcal{L}_s^{(1)}} + \frac{1}{2} \frac{\mathcal{L}_a^{(e)}}{\mathcal{L}_a^{(1)}}, \quad (2)$$

where $\mathcal{L}_s^{(e)}$ and $\mathcal{L}_a^{(e)}$ are the epoch- e validation losses for the sarcasm and MGTD tasks on their respective validation sets. Normalization reduces the influence of differences in task-specific loss scales and learning dynamics.

Five runs are defined by the SarcTrain validation folds: four folds are used for sarcasm training and one for validation, while AuthTrain uses a fixed 90/10 split. Each run reloads the pre-trained encoder and randomly initializes new modules and heads. Configurations share fold assign-

ments and nominal seeds, enabling split-matched comparisons but not identical initialization.

DualTest is not included in any training or validation split for either task. Five-run standard deviations combine fold and initialization variation. Wilson intervals are conditional on a fixed final model and cover test-instance, not training or generator, uncertainty (Table 3; Appendix C.7).

For each configuration used in the subgroup analysis, a final model is trained on all applicable data with a fixed epoch count determined from the five validation runs, an independent seed, and no early stopping. DualTest is not used to train these models or determine their training duration, and the final models are not included in five-run aggregates; §6 uses the final RCNN dual-head model (Appendix B).

5 Experiments and Diagnosis

With the sarcasm and MGTD models fixed, we use DualTest first to compare encoder backbones and intermediate configurations, then to measure the change between the joint- and single-task procedures, and finally to partition predictions into source $\times$ sarcasm groups. Aggregate results are five-run means $\pm$ standard deviations; fixed final models support paired instance-level subgroup comparisons and Wilson intervals.

5.1 Backbone Comparison

We compare dual-head mBERT (Devlin et al., 2019), XLM-RoBERTa-base (Conneau et al., 2020), DeBERTa-v3-base, and DeBERTa-v3-large (He et al., 2023) under one evaluation protocol, adding the two intermediate variants on the large encoder. Positive-class F1 is computed for the sarcastic class in sarcasm detection and the human class in MGTD over the full DualTest set.

Table 1 shows higher mean F1 for the DeBERTa configurations than for mBERT and XLM-R. The RCNN configuration has the highest means, reaching 0.6309 sarcasm F1 and 0.9301 MGTD F1. Across the five runs, the two intermediate variants also have 34–47% lower standard deviations than the plain large model, providing configuration-level evidence of lower run-to-run variation. Because backbone identity, scale, and intermediate representation differ jointly, these results support only configuration-level comparisons. For the separately trained final RCNN model, sarcasm F1 is 0.819 in Chinese and 0.429 in English; prevalence,

Model	Sarc. F1	MGTD F1
mBERT dual-head	0.4945 $\pm$.036	0.8743 $\pm$.006
XLM-RoBERTa-base	0.4603 $\pm$.057	0.8321 $\pm$.025
dual-head		
DeBERTa-v3-base	0.5646 $\pm$.056	0.8952 $\pm$.004
head		
DeBERTa-v3-large	0.6126 $\pm$.031	0.9221 $\pm$.010
dual-head		
+ interaction projection	0.6076 $\pm$.020	0.9126 $\pm$.007
+ RCNN	0.6309 $\pm$.020	0.9301 $\pm$.005

Table 1: Descriptive dual-head results across backbones and architectural variants (DualTest; five-run mean $\pm$ standard deviation; bold indicates the highest mean).

Arch.	Task	Dual	Single	Δ
Plain	Sarc.	0.6126 $\pm$.031	0.6734 $\pm$.010	−6.08
Plain	MGTD	0.9221 $\pm$.010	0.9054 $\pm$.027	+1.67
RCNN	Sarc.	0.6309 $\pm$.020	0.6751 $\pm$.022	−4.43
RCNN	MGTD	0.9301 $\pm$.005	0.9253 $\pm$.006	+0.49

Table 2: Dual-head vs. single-head models (mean $\pm$ standard deviation over five fold-based runs; Δ = dual − single, in percentage points).

genre, annotation, and language may all contribute to this difference, so it should not be read as a pure language effect. Appendix C provides macro-averaged, human-subset, and language-stratified results.

5.2 Dual-Head versus Single-Head Comparison

Within each architecture, dual- and single-head models share the encoder, intermediate structure, sarcasm data, fold assignments, and sarcasm-specific hyperparameters; dual runs additionally receive AuthTrain batches and MGTD updates. Single-head controls process the full sarcasm training fold each epoch, whereas dual-head models draw sarcasm batches under the 2:1 schedule (§4.2).

Joint training lowers mean sarcasm F1 by 6.08 pp for the plain model and 4.43 pp for RCNN, whereas mean MGTD F1 changes by only +1.67 and +0.49 pp. In all ten split-matched comparisons—five per architecture—sarcasm F1 is lower for the jointly trained model, with differences from −0.8 to −8.6 pp. Under these metrics, the degradation is therefore specific to sarcasm detection. We next identify which source and sarcasm groups account for it. RCNN shows a smaller dual-single gap, motivating capacity- and compute-matched tests of which RCNN compo-

nents contribute to this difference.

5.3 Source $\times$ Sarcasm Subgroup Analysis

An overall F1 score can hide a failure concentrated in one type of text. We therefore divide DualTest into four groups defined by source and sarcasm label. Human labels are dataset annotations, whereas AI labels record the prompted condition. Because each group contains only one reference class, we report recall for sarcastic texts and false-positive rate for non-sarcastic texts.

5.3.1 Three Architectures and Single-Head Controls

The three dual-head variants show whether the subgroup error changes with the intermediate representation. Plain and RCNN single-head models provide same-architecture controls. Within each pair, both models use the same all-human sarcasm data; only the dual-head model is additionally trained on MGTD. Their difference therefore measures the complete joint-training procedure. [Table 3](#) reports sarcasm predictions; corresponding MGTD results are in [Appendix C](#).

False-positive rates on AI $\times$ non-sarcastic instances are 86.3% for the plain dual-head model, 40.9% for interaction projection, 34.1% for RCNN, and 19.3% for both same-architecture single-head controls; the intermediate rates are therefore more than 45 pp below the plain rate.

On the same AI non-sarcastic texts, 1,021 false positives occur only under the plain dual-head model, versus 7 only under its single-head control ($p < .001$). The control receives the same all-human sarcasm supervision but is not jointly trained on MGTD, tying the large increase to the joint-training procedure under comparison. RCNN shows the same direction with a smaller gap. The plain and RCNN models provide the matched contrasts, while interaction projection adds complementary configuration-level evidence.

No analogous extreme subgroup error appears in the MGTD head: its false-positive rate is 6.1% for plain and 6.6% for RCNN. Within AI-generated texts, sarcasm AUC remains 0.82–0.88 across models, so the relative ranking of sarcastic and non-sarcastic AI texts is largely preserved. Joint training instead shifts predicted sarcasm probabilities upward, causing many texts prompted to be non-sarcastic to cross the fixed 0.5 threshold. The failure therefore takes the form of an AI-associated

score increase in this subgroup, not a complete loss of within-AI ranking ability.

The matched plain comparison makes the concentration especially clear: its non-sarcastic false-positive rate changes by 4.2 pp on human text but by 67.0 pp on AI-generated text. The aggregate F1 decrease is therefore driven by a sharply localized source $\times$ class effect rather than a uniform loss across DualTest.

5.3.2 Mechanism Hypothesis and Representation-Level Diagnostics

We next ask whether source information in the shared representation is associated with the higher sarcasm scores. A linear probe predicts source from both dual-head (AUC 0.986) and same-architecture single-head (0.974) representations. The comparable single-head AUC shows that source decodability is a background property shared by both models. In the dual-head model, texts with a higher MGTD-head AI logit also tend to have a higher sarcasm logit in the target subgroup ($\rho = 0.519$). After covariance normalization, the source directions estimated from the MGTD head and the probe are also moderately aligned with the sarcasm direction. Together, the logit correlation and directional alignment provide correlational evidence linking source-related structure to the affected sarcasm decisions.

We then modify the frozen representations and recompute predictions. LEACE ([Belrose et al., 2023](#)), which reduces linearly decodable source information, lowers source-probe AUC from 0.982 to 0.589. It also changes the AI/human non-sarcastic false-positive rates from 84.8%/32.8% to 68.7%/49.5%. The nearly unchanged unweighted mean (58.8% to 59.1%) indicates that LEACE reallocates errors between source groups without reducing their unweighted mean.

A key control removes a high-variance direction constructed to be orthogonal to the estimated linear source direction. This changes the AI/human false-positive rates to 66.7%/54.1% even though measured source decodability remains 0.982; ten sampled random directions show no comparable change. Together, these controls show prediction sensitivity to a non-random high-variance direction. The source-correlated shortcut hypothesis therefore provides a focused mechanism for further analysis, with the semantic content of the effective direction as the next causal target ([Appendix E](#)).

Model	H sarc. rec.	H non-sarc. FP	AI sarc. rec.	AI non-sarc. FP
Plain dual-head	67.1%	32.9%	98.2%	86.3% [84.5, 88.0]
Interaction-projection dual-head	71.9%	37.6%	94.4%	40.9% [38.5, 43.4]
RCNN dual-head	80.8%	40.5%	89.3%	34.1% [31.8, 36.5]
Plain single-head (sarcasm only)	67.8%	28.7%	76.6%	19.3% [17.4, 21.4]
RCNN single-head (sarcasm only)	66.2%	28.5%	79.1%	19.3% [17.4, 21.4]

Table 3: Five sarcasm classifiers on four source $\times$ sarcasm quadrants (recall / false-positive rate, FP). Human labels are annotations; AI labels record prompted conditions. Brackets give model-conditional Wilson 95% intervals; full intervals and paired tests are in [Appendix C.7](#).

5.3.3 Cross-Generator Replication

We keep the trained detectors fixed and compare the DualTest AI texts generated by deepseek-r1:8b with replacements generated by Qwen3.5-9B (Ollama tag qwen3.5:9b) (Qwen Team, 2026). After quality control, the non-sarcastic subsets generated by deepseek-r1:8b and Qwen3.5-9B contain 1,513 and 1,550 outputs. The plain dual-head false-positive rates are 86.3% and 91.3%, whereas the corresponding single-head rates are 19.3% and 7.9%; the dual-single gap therefore changes from 67.0 to 83.4 pp. On the Qwen3.5-generated subset, the RCNN and interaction-projection rates remain much lower than the plain dual-head rate, at 22.1% and 29.5%. The gap has the same direction and the configurations have the same ordering with Qwen3.5, although the magnitudes differ (Appendix C).

6 Exploratory Comparison of Detector Scores and LLM-Judge Ratings

As a downstream test, we compare the RCNN detector’s human-authorship and sarcasm probabilities with source-blinded LLM ratings of human-likeness and sarcasm. We evaluate 600 outputs from Qwen3-8B (Yang et al., 2025) before and after QLoRA (Detters et al., 2023) fine-tuning on GenTrain. DeepSeek-V3 (DeepSeek-AI, 2024), Gemini-2.5-Flash-Lite (Google DeepMind, 2025), and GPT-5-nano (OpenAI, 2025) rate outputs from 180 Chinese topic-comment and 120 English rewriting prompts; language is therefore coupled with generation task. Because human-authorship probability and perceived human-likeness are distinct constructs, this comparison tests whether classifier scores can serve as generation-evaluation proxies.

Detector probabilities and judge ratings do not move consistently. In Chinese, SFT raises the RCNN detector’s sarcasm score from 0.780 to 0.895 but lowers the mean LLM sarcasm rating

from 3.99 to 3.80; the detector-judge correlation is -0.020 in Chinese and $+0.338$ in English. SFT also raises human-authorship probabilities in both task-language settings while mean perceived human-likeness ratings decline, yielding a pooled correlation of -0.212 ; weaker within-condition correlations show that this value is driven mainly by the condition-level reversal.

The sarcasm-detector change is setting-dependent ($+0.115$ in Chinese; -0.124 in English), whereas SFT receives lower mean sarcasm ratings in all six language $\times$ judge comparisons, five remaining significant after FDR correction. Detector probabilities can therefore remain useful classification outputs without preserving the ordering or condition-level direction required of style or quality scores.

Individual-text judge agreement is low to moderate; a plain-detector control changes correlations by at most 0.02. The human-authorship/human-likeness reversal and judge-specific sarcasm results motivate external validation before detector probabilities are reused as quality scores or rewards; [Appendices C–D](#) report full protocols, tests, and cases.

7 Conclusion

Our results expose a source-related shortcut risk in joint MGTD-sarcasm training: relative to a same-architecture single-head control, the plain dual-head model raises the false-positive rate on AI text prompted to be non-sarcastic by 67.0 pp. The pattern holds across languages and with Qwen3.5-9B, while RCNN and interaction projection yield lower rates. Representation interventions show that source-related and other high-variance directions shape source-group errors, providing targets for causal analysis. Human-authorship probabilities and human-likeness ratings diverge, requiring external validation of detector scores by source and class before use as quality metrics or rewards.

Limitations

DualTest combines human sarcasm annotations with instruction-conditioned labels for AI texts, so AI-side false positives are defined relative to requested style rather than independent perceived-sarcasm annotation. Generation failures from `deepseek-r1:8b` leave 80 human texts without generated counterparts. Because oversampling precedes fold assignment, some SarcTrain copies occur in different folds and can therefore appear on both sides of a training-validation split, introducing dependence into checkpoint selection. The affected records are absent from DualTest.

The core dual-single comparisons evaluate the complete joint-training procedure: architecture, sarcasm data, folds, sarcasm-head learning rate, and class weights are shared, whereas MGTD batches, task scheduling, shared updates, and joint early stopping are introduced together. Computed-matched controls could further decompose this effect. Cross-architecture results are configuration-level because capacity and optimization differ, and the cost-sensitive loss can affect absolute rates. AuthTrain is generated with `deepseek-r1:8b`; Qwen3.5-9B replaces only DualTest’s AI side and therefore provides a directional replication on test data from a second generator rather than a retraining-based replication.

The fixed-model linear analyses connect source-related structure to behavior but do not isolate a unique causal feature or characterize variation across training seeds. The detector-judge study uses ratings from LLMs rather than humans and couples language with generation task. These boundaries motivate independent style annotation, matched-budget controls, cross-seed mechanism tests, and broader generator and human-evaluation studies.

Ethical considerations

All human-authored texts come from publicly released research datasets: iSarcasm, CSC, ToSarcasm, News Headlines, and the Open Chinese Internet Sarcasm Corpus. Explicit repository licenses were available for iSarcasm and CSC; for sources without an identified redistribution license, we do not redistribute the human-authored text (Appendix A). We collect no new user data. We did not conduct any additional audit of the released corpora for personally identifiable information or offensive content.

All AI texts were generated locally using publicly released open-weight models (`deepseek-r1:8b`, Qwen3.5-9B, and the base and QLoRA-adapted Qwen3-8B). They are used only for training and evaluation and are labeled as AI-generated. No new human participants were recruited; the study analyzes human-authored texts from previously released datasets. The automated evaluation transmitted each evaluation input and generated output, together with rating instructions, to the DeepSeek-V3, Gemini-2.5-Flash-Lite, and GPT-5-nano APIs; it did not transmit dataset metadata or ground-truth source labels.

The detectors were developed for research diagnosis. Their outputs should neither be used as evidence of authorship or misconduct at the individual-text level without human review and domain-specific validation nor be deployed for content moderation or automated sanctions without evaluating out-of-distribution behavior. The class-conditional failure diagnosed here is an example of systematic subgroup error affecting a particular type of generated content.

AI-assisted tools were used for grammar correction, wording suggestions, and formatting. They did not generate the reported experimental outputs or statistics; the authors determined the study design, analysis, and conclusions and take full responsibility for the work.

References

- Ibrahim Abu Farha, Silviu Vlad Oprea, Steven Wilson, and Walid Magdy. 2022. [SemEval-2022 task 6: iSarcasmEval, intended sarcasm detection in English and Arabic](#). In *Proceedings of the 16th International Workshop on Semantic Evaluation (SemEval-2022)*.
- Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. 2023. [LEACE: Perfect linear concept erasure in closed form](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Yoav Benjamini and Yosef Hochberg. 1995. [Controlling the false discovery rate: A practical and powerful approach to multiple testing](#). *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300.
- Jane Bevendorff, Matti Wiegmann, Emmelie Richter, Martin Potthast, and Benno Stein. 2025. [The two paradigms of LLM detection: Authorship attribution vs. authorship verification](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 3762–3787.

- Silvia Casola, Simona Frenda, Soda Maren Lo, Erhan Sezerer, Antonio Uva, Valerio Basile, Cristina Bosco, Alessandro Pedrani, Chiara Rubagotti, Viviana Patti, and Davide Bernardi. 2024. [MultiPICO: Multilingual perspectivist irony corpus](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 16008–16021.
- Junyi Chai, Shenyu Lu, and Xiaoqian Wang. 2025. [Identifying and mitigating spurious correlation in multi-task learning](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 25698–25707.
- Rhys Compton, Lily Zhang, Aahlad Puli, and Rajesh Ranganath. 2023. [When more is less: Incorporating additional datasets can hurt performance by introducing spurious correlations](#). In *Proceedings of the 8th Machine Learning for Healthcare Conference*, volume 219 of *Proceedings of Machine Learning Research*, pages 110–127. PMLR.
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. [Unsupervised cross-lingual representation learning at scale](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*.
- DeepSeek-AI. 2024. [DeepSeek-V3 technical report](#). *arXiv preprint arXiv:2412.19437*.
- DeepSeek-AI. 2025. [DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning](#). *arXiv preprint arXiv:2501.12948*.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. [QLoRA: Efficient finetuning of quantized LLMs](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: Pre-training of deep bidirectional transformers for language understanding](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*.
- Xiyang Du, Dou Hu, Jin Zhi, Lianxin Jiang, and Xiaofeng Shi. 2022. [PALI-NLP at SemEval-2022 task 6: iSarcasmEval- fine-tuning the pre-trained model for detecting intended sarcasm](#). In *Proceedings of the 16th International Workshop on Semantic Evaluation (SemEval-2022)*, pages 815–819.
- Liam Dugan, Alyssa Hwang, Filip Trhlik, Andrew Zhu, Josh Magnus Ludan, Hainiu Xu, Daphne Ippolito, and Chris Callison-Burch. 2024. [RAID: A shared benchmark for robust evaluation of machine-generated text detectors](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 12463–12492.
- Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. 2021. [Amnesic probing: Behavioral explanation with amnesic counterfactuals](#). *Transactions of the Association for Computational Linguistics (TACL)*, 9:160–175.
- Leo Gao, John Schulman, and Jacob Hilton. 2023. [Scaling laws for reward model overoptimization](#). In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, pages 10835–10866.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. [Shortcut learning in deep neural networks](#). *Nature Machine Intelligence*, 2:665–673.
- Aniruddha Ghosh and Tony Veale. 2016. [Fracking sarcasm using neural network](#). In *Proceedings of the 7th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis (WASSA)*.
- Google DeepMind. 2025. [Gemini 2.5 flash-lite model card](#). Published September 26, 2025.
- Biyang Guo, Xin Zhang, Ziyuan Wang, Minqi Jiang, Jinran Nie, Yuxuan Ding, Jianwei Yue, and Yupeng Wu. 2023. [How close is ChatGPT to human experts? comparison corpus, evaluation, and detection](#). *arXiv preprint arXiv:2301.07597*.
- Han He and Jinho D. Choi. 2021. [The stem cell hypothesis: Dilemma behind multi-task learning with transformer encoders](#). In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Pengcheng He, Jianfeng Gao, and Weizhu Chen. 2023. [DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing](#). In *International Conference on Learning Representations (ICLR)*.
- John Hewitt and Percy Liang. 2019. [Designing and interpreting probes with control tasks](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*.
- Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. 2020. [The curious case of neural text degeneration](#). In *International Conference on Learning Representations (ICLR)*.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022a. [LoRA: Low-rank adaptation of large language models](#). In *International Conference on Learning Representations (ICLR)*.
- Ziniu Hu, Zhe Zhao, Xinyang Yi, Tiansheng Yao, Lichan Hong, Yizhou Sun, and Ed H. Chi. 2022b. [Improving multi-task generalization via regularizing spurious correlation](#). In *Advances in Neural Information Processing Systems (NeurIPS)*.

907	Hyewon Jang, Bettina Braun, and Diego Frassinelli.	DetectGPT: Zero-shot machine-generated text detec-	961
908	2023. Intended and perceived sarcasm between	tion using probability curvature. In <i>Proceedings</i>	962
909	close friends: What triggers sarcasm and what gets	of the 40th International Conference on Machine	963
910	conveyed? In <i>Proceedings of the Annual Meeting of</i>	Learning (ICML).	964
911	<i>the Cognitive Science Society</i> , volume 45.		
912	Hyewon Jang and Diego Frassinelli. 2024. Generaliz-	David Mueller, Nicholas Andrews, and Mark Dredze.	965
913	able sarcasm detection is just around the corner, of	2022. Do text-to-text multi-task learners suffer from	966
914	course! In <i>Proceedings of the 2024 Conference of</i>	task conflict? In <i>Findings of the Association for</i>	967
915	<i>the North American Chapter of the Association for</i>	<i>Computational Linguistics: EMNLP 2022.</i>	968
916	<i>Computational Linguistics: Human Language Tech-</i>		
917	<i>nologies (NAACL-HLT)</i> , pages 4238–4249.	Ollama. 2025. DeepSeek-R1-0528-Qwen3-8B model	969
		card. Ollama deployment tag <code>deepseek-r1:8b</code> ; ac-	970
		cessed 2026-07-21.	971
918	Rabeeh Karimi Mahabadi, Sebastian Ruder, Mostafa	OpenAI. 2025. OpenAI GPT-5 system card. August	972
919	Dehghani, and James Henderson. 2021. Parameter-	2025.	973
920	efficient multi-task fine-tuning for transformers via		
921	shared hypernetworks. In <i>Proceedings of the 59th</i>	Silviu Oprea and Walid Magdy. 2020. iSarcasm: A	974
922	<i>Annual Meeting of the Association for Computa-</i>	dataset of intended sarcasm. In <i>Proceedings of the</i>	975
923	<i>tional Linguistics (ACL).</i>	<i>58th Annual Meeting of the Association for Compu-</i>	976
924	Akshay Khatry and Pranav P. 2020. Sarcasm detection	<i>tational Linguistics (ACL).</i>	977
925	in tweets with BERT and GloVe embeddings. In <i>Pro-</i>		
926	<i>ceedings of the Second Workshop on Figurative Lan-</i>	Jonas Pfeiffer, Aishwarya Kamath, Andreas Rücklé,	978
927	<i>guage Processing (FigLang).</i>	Kyunghyun Cho, and Iryna Gurevych. 2021. AdapterFusion:	979
928	Siwei Lai, Liheng Xu, Kang Liu, and Jun Zhao. 2015.	Non-destructive task composi-	980
929	Recurrent convolutional neural networks for text	tion for transfer learning. In <i>Proceedings of the</i>	981
930	classification. In <i>Proceedings of the AAAI Confer-</i>	<i>16th Conference of the European Chapter of the</i>	982
931	<i>ence on Artificial Intelligence</i> , volume 29, pages	<i>Association for Computational Linguistics (EACL).</i>	983
932	2267–2273.		
933	Olivier Ledoit and Michael Wolf. 2004. A well-	Jonas Pfeiffer, Ivan Vulić, Iryna Gurevych, and Sebas-	984
934	conditioned estimator for large-dimensional covari-	tian Ruder. 2020. MAD-X: An adapter-based frame-	985
935	ance matrices. <i>Journal of Multivariate Analysis</i> ,	work for multi-task cross-lingual transfer. In <i>Pro-</i>	986
936	88(2):365–411.	<i>ceedings of the 2020 Conference on Empirical Meth-</i>	987
		<i>ods in Natural Language Processing (EMNLP).</i>	988
937	Chong Li, Shaonan Wang, Yunhao Zhang, Jiajun	Jonathan Pilault, Amine El hattami, and Christopher	989
938	Zhang, and Chengqing Zong. 2023. Interpreting and	Pal. 2021. Conditionally adaptive multi-task learn-	990
939	exploiting functional specialization in multi-head at-	ing: Improving transfer learning in NLP using fewer	991
940	tention under multi-task learning. In <i>Proceedings of</i>	parameters and less data. In <i>International Confer-</i>	992
941	<i>the 2023 Conference on Empirical Methods in Natu-</i>	<i>ence on Learning Representations (ICLR).</i>	993
942	<i>ral Language Processing (EMNLP)</i> , pages 16460–		
943	16476.	Qwen Team. 2026. Qwen3.5: Towards native multi-	994
944	Bin Liang, Zijie Lin, Bing Qin, and Ruifeng Xu. 2022.	modal agents. Qwen3.5-9B model weights under	995
945	Topic-oriented sarcasm detection: New task, new	Apache 2.0; accessed 2026-07-21.	996
946	dataset and new method. In <i>Proceedings of the 21st</i>		
947	<i>Chinese National Conference on Computational Lin-</i>	Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael	997
948	<i>guistics (CCL)</i> , pages 557–568.	Twiton, and Yoav Goldberg. 2020. Null it out:	998
949	Bo Liu, Xingchao Liu, Xiaojie Jin, Peter Stone, and	Guarding protected attributes by iterative nullspace	999
950	Qiang Liu. 2021. Conflict-averse gradient descent	projection. In <i>Proceedings of the 58th Annual Meet-</i>	1000
951	for multi-task learning. In <i>Advances in Neural Infor-</i>	<i>ing of the Association for Computational Linguistics</i>	1001
952	<i>mation Processing Systems (NeurIPS).</i>	<i>(ACL).</i>	1002
953	Rishabh Misra and Prahal Arora. 2023. Sarcasm detec-	Ellen Riloff, Ashequl Qadir, Prafulla Surve, Lalindra	1003
954	tion using news headlines dataset. <i>AI Open</i> , 4:13–	De Silva, Nathan Gilbert, and Ruihong Huang. 2013.	1004
955	18.	Sarcasm as contrast between a positive sentiment	1005
956	Rishabh Misra and Jigyasa Grover. 2021. Sculpting	and negative situation. In <i>Proceedings of the 2013</i>	1006
957	Data for ML: The first act of Machine Learning. In-	<i>Conference on Empirical Methods in Natural Lan-</i>	1007
958	dependently published.	<i>guage Processing (EMNLP).</i>	1008
959	Eric Mitchell, Yoonho Lee, Alexander Khazatsky,	John Schulman, Filip Wolski, Prafulla Dhariwal, Alec	1009
960	Christopher D. Manning, and Chelsea Finn. 2023.	Radford, and Oleg Klimov. 2017. Proximal	1010
		policy optimization algorithms. <i>arXiv preprint</i>	1011
		<i>arXiv:1707.06347.</i>	1012

1013	Asa Cooper Stickland and Iain Murray. 2019. BERT and PALs: Projected attention layers for efficient adaptation in multi-task learning . In <i>Proceedings of the 36th International Conference on Machine Learning (ICML)</i> .	1068
1014		1069
1015		1070
1016		1071
1017		1072
1018	Yi Tay, Luu Anh Tuan, Siu Cheung Hui, and Jian Su. 2018. Reasoning with sarcasm by reading in-between . In <i>Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)</i> .	1073
1019		1074
1020		1075
1021		1076
1022		1077
1023	Bertie Vidgen, Alex Harris, Dong Nguyen, Rebekah Tromble, Scott Hale, and Helen Margetts. 2019. Challenges and frontiers in abusive content detection . In <i>Proceedings of the Third Workshop on Abusive Language Online</i> , pages 80–93.	1078
1024		1079
1025		1080
1026		1081
1027		
1028	Changli Wang, Fang Yin, Jiafeng Liu, and Rui Wu. 2026. ViSP: A PPO-enhanced framework for multi-modal sarcasm generation with contrastive learning . <i>Neurocomputing</i> , 678:133185.	1082
1029		1083
1030		1084
1031		1085
1032		1086
1033	Yuxia Wang, Jonibek Mansurov, Petar Ivanov, Jinyan Su, Artem Shelmanov, Akim Tsvigun, Chenxi Whitehouse, Osama Mohammed Afzal, Tarek Mahmoud, Toru Sasaki, Thomas Arnold, Alham Fikri Aji, Nizar Habash, Iryna Gurevych, and Preslav Nakov. 2024. M4: Multi-generator, multi-domain, and multi-lingual black-box machine-generated text detection . In <i>Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)</i> , pages 1369–1407.	1087
1034		1088
1035		1089
1036		1090
1037		1091
1038		1092
1039		
1040		1093
1041		1094
1042		1095
1043	Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models . In <i>Advances in Neural Information Processing Systems (NeurIPS)</i> , pages 24824–24837.	1096
1044		1097
1045		1098
1046		1099
1047		
1048		1100
1049	Changsong Wen, Guoli Jia, and Jufeng Yang. 2023. DIP: Dual incongruity perceiving network for sarcasm detection . In <i>Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)</i> , pages 2540–2550.	1101
1050		1102
1051		1103
1052		1104
1053		1105
1054		1106
1055	Lianwei Wu, Yuan Rao, Haolin Jin, Ambreen Nazir, and Ling Sun. 2019. Different absorption from the same sharing: Sifted multi-task learning for fake news detection . In <i>Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)</i> .	1107
1056		1108
1057		1109
1058		
1059		1110
1060		1111
1061		1112
1062	Sen Wu, Hongyang R. Zhang, and Christopher Ré. 2020. Understanding and improving information transfer in multi-task learning . In <i>International Conference on Learning Representations (ICLR)</i> .	1113
1063		1114
1064		1115
1065		1116
1066	An Yang et al. 2025. Qwen3 technical report . <i>arXiv preprint arXiv:2505.09388</i> .	1117
1067		1118
		1119
		1120
		1121
	Linshuo Yang and Daisuke Ikeda. 2024. The influence of linguistic attribute differences in multilingual datasets on sarcasm detection . <i>International Journal of Service and Knowledge Management</i> , 8(2):1–15.	
	Wenqian Ye, Luyang Jiang, Eric Xie, Guangtao Zheng, Yunsheng Ma, Xu Cao, Dongliang Guo, Daiqing Qi, Zeyu He, Yijun Tian, Christopher W. Porter, Megan Coffee, Zhe Zeng, Sheng Li, Ziran Wang, Ting-Hao (Kenneth) Huang, James Matthew Rehg, Henry Kautz, and Aidong Zhang. 2026. The clever Hans mirage: A comprehensive survey on spurious correlations in machine learning . <i>Transactions on Machine Learning Research</i> .	
	Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. 2020. Gradient surgery for multi-task learning . In <i>Advances in Neural Information Processing Systems (NeurIPS)</i> .	
	Tan Yue, Xuzhao Shi, Rui Mao, Zonghai Hu, and Erik Cambria. 2024. SarcNet: A multilingual multimodal sarcasm detection dataset . In <i>Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)</i> , pages 14325–14335.	
	Zhihan Zhang, Wenhao Yu, Mengxia Yu, Zhichun Guo, and Meng Jiang. 2023. A survey of multi-task learning in natural language processing: Regarding task relatedness and training methods . In <i>Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)</i> , pages 943–956.	
	Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena . In <i>Advances in Neural Information Processing Systems (NeurIPS)</i> .	
	Yizhang Zhu. 2022. Open Chinese internet sarcasm corpus construction: An approach . <i>Frontiers in Computing and Intelligent Systems</i> , 2(1):7–9.	

A Detailed Dataset Statistics and Quality Checks

A.1 Corpus details. The pooled collection contains 44,862 instances in total (86.9% English / 13.1% Chinese). Chinese–English 1:1 balance in SarcTrain is achieved through bucketed sampling. The two English buckets and the Chinese non-sarcastic bucket contain more than 2,500 available instances and are randomly downsampled within each bucket. The Chinese sarcastic bucket contains 2,396 source records representing 2,394 unique texts; 104 records are added by

Corpus	Lang.	Inst.	Use
iSarcasm (SemEval-2022 T6)	English	3,467	Sarc. train + SFT
CSC	English	6,878	Sarc. train
ToSarcasm	zh 3,892 / en 6	3,898	Sarc. train + SFT
News Headlines v2	English	28,619	Sarc. train
Open Chinese Internet Sarcasm Corpus	Chinese	2,000	Sarc. train

Table 4: Input corpora. Genre / annotation: iSarcasm — tweets, speaker self-annotation; CSC — conversational responses; ToSarcasm — topic comments, third-party annotation; News Headlines — distant supervision (The Onion / HuffPost); Open Chinese Corpus — Weibo, third-party annotation.

oversampling to reach 2,500. When source-level text duplicates are counted, the final bucket’s text-duplication rate is 4.24% (A.3).

Licensing and release scope. The iSarcasm and CSC repositories specify the MIT License. We did not identify explicit redistribution licenses in the releases we accessed for ToSarcasm, News Headlines v2, or the Open Chinese Internet Sarcasm Corpus. We therefore do not redistribute human-authored text from those sources. Planned research artifacts are limited to eligible AI-generated text, labels and metadata, and reconstruction scripts that require users to obtain the human-authored corpora from their original sources.

Lang.	Source	Sarc.	Non-sarc.
English	Human	200	1,200
English	AI	192	1,174
Chinese	Human	623	349
Chinese	AI	587	339

Table 5: Language $\times$ source $\times$ sarcasm distribution of DualTest. Human labels are dataset annotations; AI labels record prompted conditions.

A.2 Three-dimensional distribution of Dual-Test.

A.3 Data quality checks. Automated checks confirm language and label balance across the five SarcTrain folds. AuthTrain contains 9,996 complete human–AI pairs and three unmatched human instances. Record-level matching confirms that neither SarcTrain nor AuthTrain shares records with the human or AI side of DualTest; no train–

test leakage is present.

SarcTrain’s text-duplication rate is 0.34% for English / 2.16% for Chinese. Chinese duplication is concentrated in the sarcastic bucket (4.24%; non-sarcastic bucket: 0.08%) and results from the oversampling used to fill that bucket (A.1). Because oversampling precedes fold assignment, 86 of the 104 added records have a source counterpart in another fold. Under the raw-text counting convention used for these duplication rates, 116 repeated-text groups involving 241 records are present when oversampled copies are included, and 98 groups span folds. Excluding the added copies and normalizing whitespace and case leaves 19 naturally duplicated groups involving 46 records, of which 16 span folds. These relationships are relevant to fold-based checkpoint selection but do not place the affected training or validation records into DualTest. Post-processing also checks for the explicit chain-of-thought markers and reasoning prefaces described in A.4 (Wei et al., 2022).

A.4 AI generation protocol. The AI texts in AuthTrain and the two DualTest versions are produced by locally deployed generation models through the Ollama API. The model tagged deepseek-r1:8b generates AuthTrain and the DeepSeek version of DualTest’s AI side; the Qwen3.5-9B model tagged qwen3.5:9b generates the alternate AI side used for cross-generator replication. Both models use temperature = 0.8, top- p = 0.95 (Holtzman et al., 2020), and a generation limit of 500 tokens. Generation instructions use four templates defined by language $\times$ style, with style determined by the seed instance’s sarcasm label. The templates constrain output style and length (20–80 Chinese characters; 10–50 English words). These constraints are intended to reduce, but do not eliminate, source differences in length; Appendix E separately repeats the representation correlation analysis within length tertiles. The complete Chinese sarcastic template used in the experiment is reproduced verbatim below:

【任务】生成一条讽刺风格的评论
 【主题】{seed}
 【要求】
 - 直接给出简短的评论，不要解释思路或分析
 - 风格要讽刺、幽默、夸张
 - 表达与主题相关的观点即可，不必追求严谨
 - 20–80 字为佳
 【评论】

Its pinyin transliteration is:

[rènwù] shēngchéng yī tiáo fěngcì
fēnggé de pínglùn
[zhǔtí] {seed}
[yāoqiú]
- zhíjiē gěichū jiǎnduǎn de pínglùn,
bùyào jiěshì sīlù huò fēnxī
- fēnggé yào fěngcì, yōumò, kuāzhāng
- biǎodá yǔ zhǔtí xiāngguān de guāndiǎn
jíkě, bùbù zhuīqiú yánjīn
- 20–80 zì wéi jiā
[pínglùn]

Its English translation is:

[Task] Generate a sarcastic-style
comment
[Topic] {seed}
[Requirements]
- Directly provide a short comment;
do not explain your reasoning or
analysis
- The style should be sarcastic,
humorous, and exaggerated
- It is sufficient to express a
viewpoint related to the topic;
strict rigor is not required
- Preferably 20--80 Chinese
characters
[Comment]

The Chinese non-sarcastic template changes only the style line to “The style should be normal and natural.” The English templates follow the same structure, using either “sarcastic, humorous, exaggerated” or “normal, natural,” with a length limit of 10–50 words.

Post-processing removes `<think>...</think>` segments, reasoning prefaces such as “好的” (*hǎo de* ‘okay’), “嗯” (*ń* ‘hmm’), “Okay,” “Let me,” and “First,” and Markdown markup. Outputs that still exceed the length limit or exhibit cyclic repetition after repeated generation attempts are discarded. AuthTrain retains 9,996 AI counterparts from 9,999 seeds; DualTest retains 2,292 counterparts from 2,372 seeds.

The Qwen3.5-9B replication has no generation failures, yielding 1,550 AI $\times$ non-sarcastic instances rather than 1,513. Generator-specific rates therefore use the full set of retained outputs for each generator rather than an identical seed intersection. For this run, Ollama thinking mode is disabled (`think = false`) to prevent the reasoning process from exhausting the output budget. The same post-processing removes any inline chain-of-thought from the `deepseek-r1:8b` outputs.

A.5 SFT training and generation-evaluation set. Eligibility filtering retains 5,257 positive-sarcasm

records suitable for generation tasks (2,972 English; 2,285 Chinese). After shuffling with seed 42, we sample 40% of these records, yielding 2,103 instances (1,214 English; 889 Chinese); language balancing then adds 325 oversampled Chinese records, producing GenTrain (2,428; 1,214 per language). By prompt type, it comprises 374 rewriting, 839 response, and 1,215 comment instances. Generation evaluation uses an independent set of 300 prompts (180 Chinese and 120 English): the Chinese task is to generate a sarcastic comment on a given topic, and the English task is to rewrite a normal sentence as a sarcastic one. This set is used for detector scoring and for all three source-blinded LLM evaluations in §6.

B Hyperparameters and Training Details

Group	Enc.	Inter.	Sarc.	MGTD
Plain / backbone	5e-7	—	5e-6	1e-5
dual				
RCNN dual	5e-7	5e-6	5e-6	5e-6
Interaction-projection dual	5e-7	5e-6	5e-6	5e-6
Plain single-task	5e-7	—	5e-6	1e-5
RCNN single-task	5e-7	5e-6	5e-6	1e-5

Table 6: Discriminative learning rates for the dual- and single-task configurations.

B.1 Discriminative learning rates. The sarcasm-head learning rate is fixed at 5e-6 across configurations. The MGTD-head rate is 1e-5 for the plain dual and single-task models, 5e-6 for the two intermediate dual-head variants, and 1e-5 for the RCNN MGTD single-task control. Thus, the core sarcasm dual–single comparisons retain the same sarcasm-specific rates within each architecture, whereas the RCNN MGTD comparison and cross-configuration results can also reflect optimization settings. Early stopping uses the normalized joint validation loss of Equation 2 for dual-head models and the task’s own validation loss for the single-task controls, with the same patience of five epochs in both cases.

B.2 Shared training hyperparameters. Training uses AdamW with default PyTorch betas (0.9, 0.999) and $\epsilon = 10^{-8}$, weight decay 0.01, and parameter-group learning rates in B.1. A LambdaLR schedule linearly warms up for 200 steps and then keeps the learning rate constant. Training runs for at most 10 epochs, with an early-stopping patience of 5 and gradient clipping at 1.0.

The physical batch size is 4 with gradient accumulation over 4 steps, giving an effective batch size of 16. The MGTD:sarcasm sampling ratio is 2:1, and sarcasm class weights are [1.0, 2.5]. Both choices were specified before testing. For the class-balanced SarcTrain dataset, the class weights implement a cost-sensitive objective that assigns a greater penalty to errors on sarcastic examples. The normalized joint validation loss is $0.5 \times [\text{sarcasm validation loss} / \text{first-epoch value}] + 0.5 \times [\text{MGTD validation loss} / \text{first-epoch value}]$. The single-head controls process the full sarcasm training fold each epoch, whereas the dual-head models draw sarcasm batches under the 2:1 task-mixing schedule.

Detector inputs use a maximum sequence length of 512 tokens. In the RCNN module, a width- k convolution uses padding $\lfloor k/2 \rfloor$ followed by ReLU; global max pooling is applied directly over the full convolution output without an additional padding mask.

All fold-based runs use a seed equal to 42 plus the fold index and share the same fold assignments and nominal seed scheme across architectures and ablations. MGTD data use a fixed 90/10 holdout in every fold. For each final-model configuration, the fixed epoch count is the mode of its five best epoch counts plus one, capped at 10 epochs; the resulting counts range from 3 to 10 across configurations. The final model uses all applicable training data, a seed offset of +1000, no validation set, and no early stopping. Main tables report five-run means and standard deviations rather than final-model point estimates.

Model sizes, software, and hardware environment. The encoders have approximately 178M (mBERT), 278M (XLM-RoBERTa-base), 184M (DeBERTa-v3-base), and 435M (DeBERTa-v3-large) parameters; the SFT generator Qwen3-8B has about 8B parameters, of which QLoRA updates only low-rank adapters. Detector training and inference are performed on a single NVIDIA GeForce RTX 3090 (24 GB). Across the five-fold backbone, architecture, and ablation experiments and final-model training, total compute is on the order of tens of GPU-hours. A single detector run takes from a few minutes to a few hours, and SFT takes about two hours. The software environment is PyTorch 2.9.0 (CUDA 12.8), transformers 4.57.1, and peft 0.18.0. AI texts are generated locally through Ollama (Appendix A.4); blinded

LLM evaluation uses API calls made in May 2026 (Appendix C.3).

B.3 SFT (QLoRA). We fine-tune Qwen3-8B using QLoRA (Detmeters et al., 2023), with 4-bit loading and bfloat16 computation. The LoRA configuration (Hu et al., 2022a) uses rank 64, alpha 16, and dropout 0 and targets q/k/v/o_proj plus gate/up/down_proj. The maximum sequence length is 2,048, with gradient checkpointing enabled. Training uses a learning rate of $5e-5$, three epochs, batch size 1, gradient accumulation 8, and a warmup ratio of 0.05. After language balancing, a seed-42 split holds out 15% of GenTrain for validation; validation loss selects the checkpoint.

C Complete Experimental Results

C.1 Macro-averaged F1 (supplement to Table 1). Under the same five-run protocol, the RCNN macro-averaged F1 scores computed from each run’s confusion matrix are 0.6678 ± 0.036 for sarcasm and 0.9298 ± 0.005 for MGTD. Sarcasm macro-F1 is higher than its positive-class F1 (0.6309) because the majority class has a higher F1. These values supplement rather than rerank the full positive-class comparison in §5.1.

C.2 Complete four-quadrant results for the MGTD head (including single-head controls). The reporting convention matches Table 3 (recall for human subsets; false-positive rate for AI subsets).

Model	H×s	H×n	AI×s	AI×n
Plain dual	91.4	91.5	3.6	6.1
Inter.-proj. dual	85.4	85.4	1.8	4.1
RCNN dual	94.8	93.2	5.5	6.6
Plain single	90.4	86.6	4.1	4.2
RCNN single	90.5	92.4	3.6	5.7

Table 7: MGTD-head four-quadrant results (%): recall for human (H) subsets, false-positive rate for AI subsets. s = sarcastic, n = non-sarcastic.

The single-head MGTD controls remain within single-digit percentage points of the dual-head models in every quadrant. No analogous extreme class-conditional error appears on the MGTD side, consistent with §5.3.

No comparable subgroup increase appears in the qwen3.5:9b replication set: AI × non-sarcastic false-positive rates are 0.9%, 1.3%, and 0.5% for plain, interaction projection, and RCNN; AI × sar-

castic rates are at most 0.4%; and overall MGTD F1 scores are 0.952, 0.917, and 0.966.

C.3 Blinded-evaluation protocol, inter-judge agreement, and group-level tests (§6). Each sample is submitted separately to each of the three judge APIs using a single prompt (May 2026; temperature = 0.3; GPT-5-nano uses its default decoding because it does not support this parameter). The prompt contains only the input and evaluated text, not the ground-truth source. It asks for source inference followed by 1–5 ratings of sarcasm and human-likeness. Base and SFT outputs use the same prompt and are evaluated separately.

Pairwise instance-level agreement is low to moderate. Spearman ρ ranges from 0.20 to 0.38 for sarcasm (all $p < 10^{-6}$) and from 0.03 to 0.23 for human-likeness (one pair is not significant). The conclusion in §6 therefore rests on consistent condition-level directions, not agreement in instance rankings.

Wilcoxon signed-rank tests with Benjamini–Hochberg FDR correction (Benjamini and Hochberg, 1995) show that SFT receives lower sarcasm ratings than the base model in five of six language $\times$ judge groups. The pooled correlations in Table 8 use the three-judge mean as a descriptive aggregate; because inter-judge agreement is limited, the judge-specific results remain necessary. Four instances each have one unparseable rating and retain two valid judge ratings; the aggregate uses the mean of the available ratings and therefore retains all 600 outputs.

Subset	n	Sarc. ρ	MGTD ρ
Overall	600	+0.136*	−0.212*
Chinese	360	−0.020	−0.218*
English	240	+0.338*	−0.270*

Table 8: Spearman correlations between detector probabilities and LLM-judge ratings (RCNN detector, $n = 600$ pooled over base and SFT). * significant; the other entries are not significant.

C.4 Paired tests between SFT and human references (supplementary). These tests are performed on the Chinese subset of the generation-evaluation set ($n = 180$; three GPT-5-nano ratings are missing, leaving 177). The comparisons provide judge- and dimension-specific calibration context: for DeepSeek-V3, SFT is significantly below the reference on human-likeness (−0.161, $p = 0.0025$), whereas for GPT-5-nano, SFT is sig-

Judge / lang.	n	SFT	Base	adj. p
DeepSeek-V3 / en	120	3.292	4.025	<.0001
DeepSeek-V3 / zh	180	3.878	4.044	.0008
Gemini-2.5-FL / en	120	3.750	3.992	.0008
Gemini-2.5-FL / zh	180	3.994	4.028	.2008*
GPT-5-nano / en	119	3.681	4.084	<.0001
GPT-5-nano / zh	180	3.517	3.911	<.0001

Table 9: Group-level tests for blinded sarcasm judgments (Wilcoxon + BH-FDR). * not significant. GPT-5-nano lacks one English base rating (C.3: $n = 119$) and three Chinese reference ratings (affecting C.4, not the C.3 Chinese row); tests use pairwise deletion.

nificantly above the reference on sarcasm (+0.186, $p = 0.0114$). The remaining four comparisons do not reach significance at the reported threshold. Taken together, the results indicate that proximity to the human reference depends on the judge and evaluated dimension.

C.5 Plain-detector control. With the plain detector, the Chinese sarcasm score changes from 0.765 to 0.763 (−0.001), the English sarcasm score changes from 0.843 to 0.659 (−0.184), and the MGTD score rises by +0.760 in Chinese and +0.705 in English. The two detectors therefore agree on the rise in MGTD scores, while the sarcasm direction depends on the detector.

Correlations are +0.142 for sarcasm and −0.214 for MGTD overall; −0.037 (not significant) and −0.209 in Chinese; and +0.342 and −0.283 in English. Each differs from the RCNN-detector value by at most 0.02. Negative associations between human-authorship probabilities and human-likeness ratings are similar across the two detectors, whereas the condition-level sarcasm-score change is architecture-dependent, especially in Chinese.

Within-condition correlations (RCNN detector). For MGTD, the base condition has $\rho = +0.03$ ($n = 300$, not significant; Chinese +0.10, English −0.09). The SFT condition has $\rho = −0.12$ ($p = 0.039$; Chinese −0.19, $p = 0.009$; English −0.12, not significant). The pooled value of −0.212 is driven mainly by the opposing condition-level changes: SFT raises the detector score by about 0.67 while lowering the mean blinded rating. There is no significant positive detector–judge alignment on this dimension within conditions.

For sarcasm, within-condition correlations are

+0.05 for Chinese base (not significant), +0.15 for Chinese SFT ($p = 0.048$), +0.41 for English base, and +0.37 for English SFT (both English values significant). Chinese alignment is therefore at most weakly positive, consistent with its near-zero pooled result.

C.6 Detector scores before and after SFT across the two task–language settings. The RCNN detector assigns the following scores to base and SFT outputs (0–1):

Lang.	Cond.	Sarc.	MGTD
Chinese	Base → SFT	0.780 → 0.895	0.289 → 0.969
English	Base → SFT	0.854 → 0.730	0.198 → 0.864

Table 10: Detector scores before and after SFT (RCNN detector).

Sarcasm scores move in opposite directions in the Chinese comment-generation and English rewriting settings: +0.115 and −0.124, respectively. GenTrain itself is balanced at 1,214 instances per language; the evaluation settings instead differ in task, prompt format, source data, and sample size, all of which may contribute to the contrast. MGTD scores rise sharply in both settings, so the detector assigns SFT outputs higher probabilities of human authorship.

The mean LLM ratings for human references are 3.77 for sarcasm and 4.17 for human-likeness; the corresponding values are 4.01 and 4.24 for the base model and 3.71 and 4.08 for SFT. From the base model to SFT, the mean human-likeness rating decreases even as the detector’s MGTD score increases, as analyzed in §6.

C.7 Conditional test-instance uncertainty: confidence intervals, paired tests, and subset F1. Conditional on each fixed final model, we quantify test-instance uncertainty with Wilson 95% intervals for quadrant rates and 95% percentile intervals from 10,000 instance-level bootstrap samples for F1. These intervals do not include uncertainty from training, generator sampling, corpus construction, or human–AI pairing. They complement the between-run standard deviations in Table 2 and Table 1. All values use the final model for each architecture and follow the reporting convention of Table 3.

The dual–single gap has the same direction on the Qwen3.5-generated AI texts, although its magnitude differs.

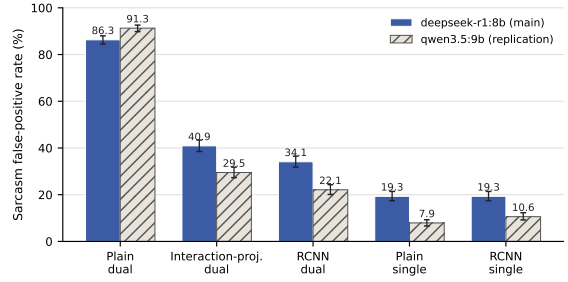


Figure 2: Sarcasm false-positive rates on the AI × non-sarcastic subset for five detectors and two generators. Error bars are Wilson 95% intervals conditional on each fixed model.

Model	DeepSeek	Qwen3.5
Plain dual	86.3 [84.5, 88.0]	91.3 [89.8, 92.6]
Inter.-proj. dual	40.9 [38.5, 43.4]	29.5 [27.3, 31.8]
RCNN dual	34.1 [31.8, 36.5]	22.1 [20.1, 24.3]
Plain single	19.3 [17.4, 21.4]	7.9 [6.6, 9.3]
RCNN single	19.3 [17.4, 21.4]	10.6 [9.2, 12.3]

Table 11: Wilson 95% intervals for the sarcasm false-positive rate on the AI × non-sarcastic subset (two generators; %).

Paired McNemar tests use predictions on the same DeepSeek-generated AI × non-sarcastic texts. The discordant-pair counts are 1,021 / 7 for plain dual vs. plain single, 239 / 15 for RCNN dual vs. RCNN single, 805 / 15 for plain dual vs. RCNN dual, and 203 / 100 for interaction projection vs. RCNN dual (all $p < .001$). The discordant counts and absolute rate differences provide the substantive effect-size information.

Model	Lang.	n	Dual pos.; rate [CI]	Single pos.; rate [CI]	Δ pp
Plain	en	1,174	987; 84.1 [81.9, 86.1]	242; 20.6 [18.4, 23.0]	+63.5
Plain	zh	339	319; 94.1 [91.1, 96.2]	50; 14.7 [11.4, 18.9]	+79.4
RCNN	en	1,174	445; 37.9 [35.2, 40.7]	247; 21.0 [18.8, 23.5]	+16.9
RCNN	zh	339	71; 20.9 [17.0, 25.6]	45; 13.3 [10.1, 17.3]	+7.7

Table 12: Language-stratified sarcasm false-positive rates on AI texts prompted to be non-sarcastic (%; Wilson 95% intervals). Exact paired McNemar tests give $p < .001$ for both plain comparisons and RCNN English, and $p = 2.4 \times 10^{-5}$ for RCNN Chinese.

In both architectures, the dual–single difference therefore has the same direction in Chinese and English. Language remains coupled with corpus, genre, and annotation setting, so these results provide stratified replication across the two task–language settings rather than an isolated language effect.

The final RCNN dual-head model’s MGTD F1 is 0.938 [0.931, 0.945] on the full set, 0.961 [0.952,

Model	Full set	Chinese	English	Human subset	Human-zh	Human-en
Plain dual-head	0.556	0.746	0.312	0.586	0.710	0.374
Interaction-projection dual-head	0.643	0.781	0.420	0.593	0.739	0.366
RCNN dual-head	0.663 [.646, .680]	0.819 [.802, .836]	0.429 [.399, .458]	0.629 [.605, .653]	0.774 [.748, .798]	0.399 [.355, .440]
Plain single-head	0.661	0.750	0.514	0.612	0.719	0.434
RCNN single-head	0.664	0.758	0.511	0.603	0.709	0.434

Table 13: Per-language and human-subset sarcasm F1 (final models; bootstrap 95% intervals included for RCNN).

0.969] in Chinese, and 0.923 [0.912, 0.933] in English. Because the human subset contains no AI-labeled negatives, a source-class F1 is not separately reported for this subset; Table 7 reports human recall directly.

Chinese sarcasm F1 is higher than English sarcasm F1 for the RCNN model (0.819 vs. 0.429). This contrast also reflects differences in prevalence, genre, and annotation procedure: sarcasm prevalence is 63.8% in Chinese but 14.2% in English, and the English human side is dominated by author-labeled intended-sarcasm tweets from iSarcasmEval whereas the Chinese side uses topic-comment-style ToSarcasm. The positive-class F1 values should therefore not be interpreted as an isolated language effect.

The separately retrained final RCNN model’s overall sarcasm F1 is 0.663, compared with 0.629 on human texts alone, because the overall score also includes AI texts. The five-run mean in Table 1 is a different estimate. Within the human subset, dual–single differences are below 0.03 and vary in direction. The negative-transfer effect is therefore concentrated in the AI-side quadrants, consistent with §5.3.

False-positive rates use a threshold of 0.5. For the plain dual-head model on AI $\times$ non-sarcastic text, the median sarcasm probability is 0.767 with an interquartile range of 0.635–0.839. The distribution therefore lies mostly above the threshold rather than placing only a few cases near it. Single-head medians are 0.204 for the plain model and 0.069 for RCNN.

AUC values within AI texts are similar across models (0.82–0.88). Joint training thus raises sarcasm scores while largely preserving their internal ordering. This pattern is consistent with a source-related component being added to the sarcasm signal.

D Illustrative Cases of Detector–LLM Score Misalignment

Illustrative cases were manually selected after inspecting the paired table of RCNN-detector scores

and ratings from the three LLMs to show large disagreements; no formal ranking or top- k selection was performed. Detector scores range from 0 to 1, and LLM scores are averaged across the three evaluators on a 1–5 scale; long texts are truncated. Chinese examples retain the original model input and provide a pinyin transliteration and an English translation in brackets.

The three groups show, respectively, high detector human-authorship scores with low LLM human-likeness ratings, high detector sarcasm scores with low LLM sarcasm ratings, and the reverse sarcasm disagreement. In the first human-likeness case, AI-generated text with obvious chain-of-thought traces is nevertheless assigned human-authorship probability 1.00 by the MGTD head, corresponding to the score disagreement discussed in §6.

L	S	$P_{\text{hum}}/\text{LLM}$	Text (excerpt)
zh	SFT	1.00 / 2.67	“好，我需要针对三星回怼华为的新闻生成一条讽刺评论。首先……” [hǎo, wǒ xūyào zhēnduì Sānxīng huíduì Huàwéi de xīnwén shēngchéng yī tiáo fěngcì pínglùn. shǒuxiān...] [“Okay, I need to generate a sarcastic comment on the news of Samsung hitting back at Huawei. First...”] (chain-of-thought leakage)
zh	SFT	1.00 / 2.33	“跑分竞赛：参数焦虑：跑分焦虑：水货：哈哈哈” [pǎofēn jìngsài: cānshù jiāolǜ: pǎofēn jiāolǜ: shuǐhuò: hāhāhā] [“Benchmark race: spec anxiety: benchmark anxiety: grey-market goods: hahaha”] (degenerate output)
en	SFT	1.00 / 2.67	“Same as me, I’m not sure” (vacuous short sentence)

Table 14: MGTD-score misalignment cases. $P_{\text{hum}} = P(\text{human})$.

Together, these selected cases motivate the hypothesis that surface markers such as mock-praise formulas, exclamation marks, and colloquial short sentences contribute to some detector decisions, while semantic reversal contributes to some LLM-judge ratings. They define candidate features for systematic tests of feature prevalence and effects

L	S	$P_{\text{sar}}/\text{LLM}$	Text (excerpt)
zh	SFT	1.00 / 2.67	“三星：我才是第一！你看看我这摄像头总分，109+96，112+89，我赢了！” [Sānxīng: wǒ cái shì dì-yī! nǐ kànkan wǒ zhè shèxiàngtóu zǒngfēn, 109+96, 112+89, wǒ yíng le!] [“Samsung: I’m the real number one! Look at my camera total scores, 109+96, 112+89, I win!”] (numbers repeated without sarcastic transformation)
zh	SFT	0.99 / 2.67	“恭喜恭喜，美国牛牛牛，继续加油” [gōngxǐ gōngxǐ, Měiguó niú niú niú, jìxù jiāyóu] [“Congratulations, congratulations, America is awesome awesome awesome, keep it up”] (formulaic mock-praise)
en	SFT	0.93 / 2.67	“2 hours left to tip it over, I’m going to the toilet”

Table 15: Detector-high / LLM-judge-low sarcasm cases. $P_{\text{sar}} = P(\text{sarcastic})$.

L	S	$P_{\text{sar}}/\text{LLM}$	Text (excerpt)
zh	Base	0.14 / 4.00	“特朗普承诺 120 亿补贴撒大饼，农民收到的却是缩水成硬币的承诺……” [Tèlǎngpǔ chéngnuò 120 yì bǔtiē xiàng sā dàbǐng, nóngmín shōudào de què shì suōshuǐ chéng yìngbì de chéngnuò...] [“Trump promised a 12-billion subsidy as if tossing out big pancakes, yet what farmers receive are promises shrunk into coins...”] (metaphorical irony)
zh	Base	0.24 / 4.00	“特斯拉：价格跳水后又跳水，消费者：钱包跳水后又跳水。” [Tèsilā: jiàgé tiàoshuǐ hòu yòu tiàoshuǐ, xiāofèizhě: qiánbāo tiàoshuǐ hòu yòu tiàoshuǐ.] [“Tesla: prices dive and then dive again; consumers: wallets dive and then dive again.”] (parallel-structure irony)
en	Base	0.12 / 4.00	“Thank you so much for this *beloved* review, Mandy! I’m *overjoyed* you enjoyed it” (asterisk emphasis)

Table 16: Detector-low / LLM-judge-high sarcasm cases. $P_{\text{sar}} = P(\text{sarcastic})$.

and provide qualitative examples of the near-zero Chinese sarcasm correlation in §6.

E Representation-Level Analysis of Source-Conditioned Error Allocation

This appendix records the complete protocol, data, and self-checks for the representation-level tests in §5.3.2. The conclusions are interpreted as described in §5.3.2.

E.1 Setup. All analyses use the frozen plain dual-head model that produces the 86.3% result reported in the main text. We use split-sample evaluation: one half is used to estimate the direction, and the other is used to compute metrics. The appendix baseline is therefore 84.8% on the evaluation half rather than 86.3% on the full set.

Implementation checks pass. Manually computed logits match model outputs with a maximum error of 1.4×10^{-6} , and the erasure-projection residual is about 10^{-6} . LEACE satisfies its defining property: after erasure, the fitting-half class-mean difference has a relative residual of 7×10^{-6} , and an in-sample linear probe is at chance (AUC 0.514). The source-probe AUC is 0.986 with true labels and 0.512 under source-label permutation.

E.2 Protocol. The linear probe is a logistic-regression classifier evaluated with five-fold cross-validation. A permuted-label control retrains the same probe after shuffling source labels (Hewitt and Liang, 2019), and correlation analyses are repeated within text-length tertiles. For alignment, Σ is the Ledoit–Wolf shrinkage covariance estimator (Ledoit and Wolf, 2004) fitted to all pooled Dual-Test representations, with eigenvalues clipped below 10^{-8} ; w denotes a task-head weight difference or probe direction. Whitened alignment is $\cos(\Sigma^{1/2}w_1, \Sigma^{1/2}w_2)$.

Erasure tests cover one direction, INLP subspaces ($k = 1\text{--}16$; Ravfogel et al., 2020), and LEACE (Belrose et al., 2023). Controls use ten sampled random whitened directions and directions estimated from permuted labels. We construct source-direction-orthogonal PC1 (srcORTH-PC1) by projecting the fitting-half representations onto the orthogonal complement of the estimated linear source direction and then extracting the maximum-variance direction in that projected space. Erasure removes each representation’s scalar projection onto this direction. This operation leaves the measured linear source decod-

ability unchanged, although orthogonality to one estimated direction does not imply independence from all multidimensional or nonlinear source information.

E.3 Diagnostic evidence. Source is highly linearly decodable: AUC is 0.986 overall, 0.990 on sarcastic texts, and 0.985 on non-sarcastic texts. The permuted-label control gives 0.512. The single-head model also yields an AUC of 0.974, showing that source decodability is a background property shared by both models. On AI $\times$ non-sarcastic texts, the MGTD-head AI logit and sarcasm-head logit correlate at $\rho = 0.519$; correlations by text-length tertile are 0.44, 0.56, and 0.63.

The sarcasm and source directions are nearly orthogonal in the original geometry ($\cos \approx -0.03$) but moderately aligned after whitening. The absolute cosine is 0.41 using the probe direction and 0.49 using the MGTD-head direction. These results provide correlational evidence consistent with the shortcut hypothesis and motivate the intervention tests below.

E.4 Intervention evidence. $\text{gap} = \text{AI} \times \text{non-sarcastic false-positive rate} - \text{human} \times \text{non-sarcastic false-positive rate}$; source AUC measures the performance of a probe retrained after erasure.

Erasure scheme	AI $\times$ n FP	H $\times$ n FP	Gap	AUC
None (baseline)	84.8	32.8	52.0	0.982
Single dir. (head / probe / random)	84.8–86.2	—	—	≈ 0.98
INLP ($k = 16$)	89.8	—	—	0.974
LEACE (linear)	68.7	49.5	19.1	0.589
Random	84.9	32.7	—	0.982
whitened $\times 10$				
srcORTH-PC1	66.7	54.1	12.6	0.982

Table 17: Intervention evidence. H $\times$ n = human $\times$ non-sarcastic; AI $\times$ n = AI $\times$ non-sarcastic; FP in %.

Single-direction and low-dimensional INLP erasure produce little change. LEACE reduces measured linear source decodability and the source-specific gap from 52.0 to 19.1; the ten sampled random-direction controls show no comparable gap change.

The orthogonal high-variance control qualifies this interpretation. It leaves the measured linear source decodability intact but changes the AI-side rate to 66.7 and the source gap to 12.6, an effect similar to LEACE. Together with the random-direction controls, this result shows that predictions are sensitive to a non-random high-variance

direction constructed after removing its projection onto the estimated source direction. The direction’s semantic content and causal role therefore remain concrete targets for further analysis.

Under LEACE, the unweighted mean false-positive rate across the two non-sarcastic source groups is almost unchanged ($58.8 \rightarrow 59.1$). This result shows that the affected directions shape how non-sarcastic errors are allocated between AI and human texts.